

JEE(Main) : ENTHUSE + LEADER COURSE (PHASE : 3)
ANSWER KEY
PART 1 : PHYSICS

SECTION-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	C	D	C	D	D	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	D	A	B	D	B	A	A	C
SECTION-B	Q.	1	2	3	4	5					
	A.	1	30	15	3	5					

PART 2 : CHEMISTRY

SECTION-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	C	A	B	D	A	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	C	B	B	D	D	D	C	A
SECTION-B	Q.	1	2	3	4	5					
	A.	1	8	1	3	5					

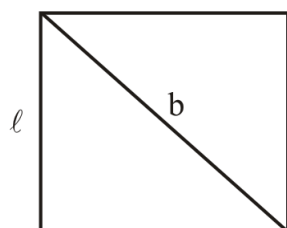
PART 3 : MATHEMATICS

SECTION-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	C	A	D	A	C	C	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	D	D	B	D	B	D	B	C
SECTION-B	Q.	1	2	3	4	5					
	A.	1	2	6	0	2					

HINT – SHEET
PART 1 : PHYSICS
SECTION-A

 1. **Ans (B)**

$$m_1 g \ell = \left(\frac{m_2 \ell^2}{3} \omega \right) + m_1 (L \omega) L$$

 2. **Ans (A)**

 $b \rightarrow \text{daigonal}$

$$(b = \ell\sqrt{2})$$

 3. **Ans (D)**

Apply WET between initial and final position :

$$\frac{1}{2} (I_1 \omega^2 + I_2 \omega^2 + I_3 \omega^2) = mg \left(\frac{\ell}{3} + \frac{2\ell}{3} + \frac{3\ell}{3} \right)$$

$$\frac{1}{2} \left(\frac{m\ell^2}{9} \omega^2 + \frac{4m\ell^2}{9} \omega^2 + m\ell^2 \omega^2 \right) = mg(2\ell)$$

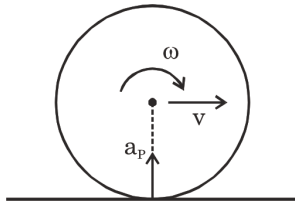
$$\frac{14}{9} m\ell^2 \omega^2 = mg4\ell$$

$$\frac{7}{18} \omega^2 \ell = g$$

$$\omega^2 = \sqrt{\frac{18g}{7\ell}}$$

$$V_B = \frac{\omega 2\ell}{3} = \sqrt{\frac{8g\ell}{7}}$$

4. Ans (C)



$$a_p = \omega^2 R = \frac{v^2}{R}$$

5. Ans (D)

$T_1 + T_2 = mg$ (Force balance)

Torque about one end of rod

$$T_2 \ell \cos 30^\circ - mg \frac{\ell}{2} \cos 30^\circ = 0$$

$$\text{So } T_1 = T_2 = \frac{mg}{2}$$

6. Ans (C)

Velocity decreases so the angular momentum.

7. Ans (D)

$$mv_0 \frac{\ell}{4} = \frac{m\ell^2}{3} \omega$$

$$\omega = \frac{3mv_0}{4\ell}$$

$$I = \frac{m\omega\ell}{2} - mv_0$$

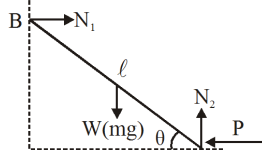
$$= \frac{3mv_0}{8} - mv_0 = -\frac{5mv_0}{8}$$

8. Ans (D)

The F.B.D. of rod is as shown.

for rod to be in translational equilibrium

$$N_1 = P \dots\dots(i)$$



$$N_2 = W = mg \dots(ii)$$

For rod to be in rotational equilibrium, net torque on rod about any axis is zero.

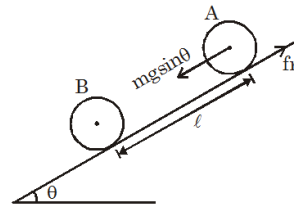
∴ Net torque on rod about B is zero i.e.,

$$mg \frac{\ell}{2} \cos \theta - N_2 \ell \cos \theta + P \ell \sin \theta = 0 \dots(iii)$$

From equation (ii) and (iii) solving we get

$$P = \frac{mg}{2} \cot \theta$$

9. Ans (B)



WET from A to B

$$\omega_{mg} + \omega_{fr} = \Delta KE.$$

$$mg \sin \theta \times \ell + w_{fr} = k$$

$$\omega_{fr} = k - mg \sin \theta \ell$$

10. Ans (B)

$$\alpha = \frac{FL/2}{\frac{ML^2}{12}} = \frac{6F}{ML}$$

$$\vec{a}_p = \vec{a}_{cm} + \vec{a}_{p,cm}$$

$$\therefore a_p = \frac{F}{M} + \frac{L}{2} \times \frac{6F}{ML} = \frac{F}{M} + \frac{3F}{M} = \frac{4F}{M}$$

11. Ans (B)

Given, slit width = a

Wavelength, $\lambda = 6500 \text{Å}$

First minimum, $\theta = 30^\circ$

For first diffraction minima,

$$n\lambda = a \sin \theta$$

$$1 \times 6500 = a \sin 30^\circ$$

$$a = 13000 \text{Å} = 1.3 \text{ micron}$$

12. Ans (B)

$$\text{The resultant intensity is } I_R = 4I_0 \cos^2 \frac{\phi}{2}$$

$$\text{and } \phi = \left(\frac{2\pi}{\lambda} \right) \times \Delta x = \frac{2\pi}{\lambda} \left[\frac{xd}{D} \right]$$

$$\therefore I_R = 4I_0 \cos^2 \left(\frac{\pi xd}{D\lambda} \right)$$

13. Ans (D)

$$\beta = \frac{D\lambda}{d} : \text{only fringe width changes}$$

* CBF remains at the same position if liquid is filled.

* If path diff at O increases, then CBF will shift upward.

PART 1 : PHYSICS

SECTION-B

14. Ans (A)

$$\Delta \Phi = \frac{2\pi}{\lambda} (\Delta x_0)$$

Optical path diff. $\Delta x_0 = \mu x$

15. Ans (B)

White spot on screen would be central maxima

where

$$Dx = 0$$

$$y = \frac{d}{2} - \frac{d}{6} = \frac{d}{3}$$

16. Ans (D)

$$RP. = \frac{\alpha}{1.22\lambda}$$

$$= \frac{6 \times 10^{-2}}{1.22 \times 540 \times 10^{-9}} = 9.1 \times 10^4$$

17. Ans (B)

$$I = \frac{1}{2} \epsilon_0 c E_0^2$$

$$E_0 = \sqrt{\frac{2I}{\epsilon_0 c}} \text{ or } \sqrt{\frac{2 \times 500 \times 10^9 \times 36\pi}{\pi \times 3 \times 10^8}}$$

$$E_0 = 2\sqrt{3} \times 10^2 \text{ N/C}$$

18. Ans (A)

Wave is propagating in -ve x-direction.

$$\therefore \hat{E} \times \hat{B} = \hat{V}$$

$$\hat{K} \times \hat{B} = -\hat{i}$$

$$\text{So } \hat{B} = \hat{j}$$

$$B_0 = \frac{E_0}{C} = \frac{1.5 \times 10^6}{3 \times 10^8} = 5 \times 10^{-3} \text{ T}$$

$$B = (B_0 \hat{j}) \sin[\omega t + kx + \phi]$$

Or

$$\vec{B} = (B_0 \hat{j}) \cos[\omega t + kx + \phi]$$

$$\vec{B} = (5 \times 10^{-3} \hat{j}) \cos(\omega t + kx)$$

19. Ans (A)

$$\text{Speed of wave} = \frac{2 \times 10^{10}}{200} = 10^8 \text{ m/s}$$

$$\text{Refractive index} = \frac{3 \times 10^8}{10^8} = 3$$

$$\text{Now refractive index} = \sqrt{\epsilon_r \mu_r}$$

$$3 = \sqrt{\epsilon_r (1)}$$

$$\Rightarrow \epsilon_r = 9$$

Option (1)

20. Ans (C)

$$i_c = i_d \text{ (always)}$$

1. Ans (1)

$$\Delta x = (\mu_{\text{rel}} - 1)t = \left[\frac{\mu_2}{\mu_1} - 1 \right] t$$

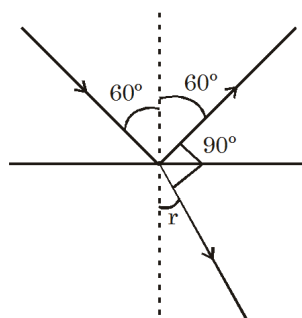
$$\mu_1 = \frac{4}{3}$$

$$\mu_2 = \frac{3}{2}$$

$$t = 8 \mu\text{m}$$

$$\Delta x = 1 \mu\text{m}$$

2. Ans (30)



At polarizing angle (Brewster's angle) angle between reflected and refracted ray is 90°

3. Ans (15)

$$2 \mu t \cos r = n\lambda$$

$$2 \times \frac{4}{3} \times t \cos 60 = 1 \times 5 \times 10^{-7}$$

$$t = 3.75 \times 10^{-7} \text{ m}$$

4. Ans (3)

$$a_1 = R\alpha - a_2 \Rightarrow 2 = 2\alpha - 4 \Rightarrow \alpha = 3$$

$$2 \text{ m/s}^2$$

5. Ans (5)

$$mg - T = ma \dots\dots(i)$$

$$T.R. = \frac{mR^2}{2} \cdot \alpha \dots\dots(ii)$$

$$a = R\alpha \dots\dots(iii)$$

$$\therefore T = \frac{mg}{3}$$

PART 2 : CHEMISTRY

SECTION-A

1. **Ans (B)**

For reversible process at equilibrium $\Delta S =$

$$\frac{q_{\text{rev}}}{T}$$

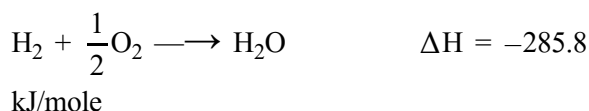
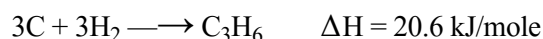
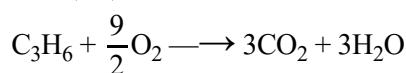
2. Ans (B)

$$\eta = \frac{T_2 - T_1}{T_2}$$

$$\eta = \frac{500 - 300}{500} = \frac{2}{5}$$

$$\eta = \frac{W_{by}}{q_{source}} \Rightarrow W_{by} = (n_C) (q_{source}) = \left(\frac{2}{5}\right)(2 \text{ kcal}) = 0.8 \text{ kcal}$$

3. **Ans (C)**



$$\begin{aligned}(\Delta H_C)_{C_3H_6} &= [3 \Delta H_{CO_2} + 3 \Delta H_{f(H_2O)} - \Delta H_{f(C_3H_6)}] \\ &= [3 \times (-394) - 3(285.8) - 20.6]\end{aligned}$$

$$(\Delta H_C)_{C_3H_6} = -2060 \text{ kJ/mole}$$

4. Ans (C)

Closed container not insulated $\Rightarrow$ can exchange energy but not matter.

5. Ans (A)

$$-w = \int P dV = \int 10 \frac{dV}{V} = 10 \ln \frac{V_2}{V_1} = 10 \times \ln 10 \text{ atm L} = 23 \text{ atm L} = 2332.2 \text{ J}$$

$$(q = \Delta U - w)$$

$$q = 420 + 2332.2 = 2752.2 \text{ J}$$

6. Ans (B)

At constant pressure

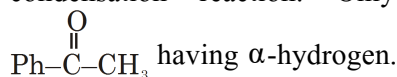
$$\Delta H = nC_p \Delta T$$

$$\Delta U = nC_v\Delta T$$

$$\text{Required function} = \frac{-\Delta w}{\Delta H} = 1 - \frac{\Delta U}{\Delta H} = 1 - \frac{C_V}{C_P} = 1 - \frac{1}{\gamma}.$$

7. Ans (D)

Carbonyl compound having α H can give aldol condensation reaction. Only acetophenone



8. **Ans (A)**

For Assertion :Acetal and ketals are basically ethers hence they must be stable in basic medium but should break down in acidic medium.

Hence assertion is correct.

For reason: Alkoxide ion ($\text{RO}^\ominus$) is not considered a good leaving group hence reason must be false

9. Ans (B)

More reactive carbonyl oxidized & cross reaction carbonyl compound will reduce.

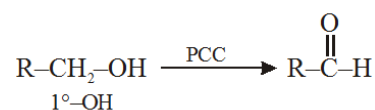
So, nitro benzaldehyde oxidized & p-methoxy benzaldehyde reduced.

10. Ans (A)

Acetaldehyde gives Iodoform, Fehling, Tollens, Benedict test

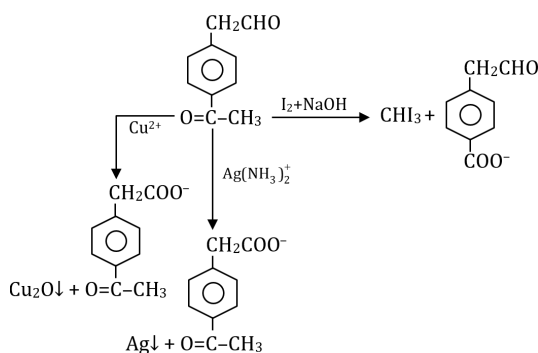
Acetone gives Iodoform test NOT Fehling,
Tollens, Benedict test

11. Ans (B)



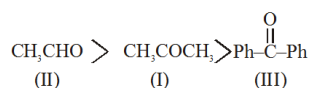
12. Ans (D)

Aldehydes give positive Tollens' and Fehling's test.

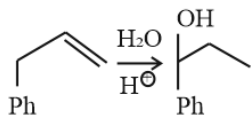


13. Ans (C)

Order of reactivity :



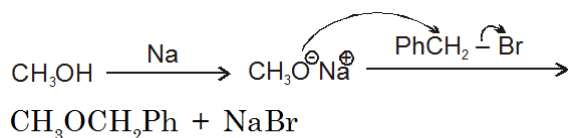
14. Ans (B)



15. Ans (B)

P.C.C. (Pyridinium chloro chromate) is mild-oxidising agent.

16. Ans (D)



Similarly for $\text{PhCH}_2\text{OH} \xrightarrow{\text{Na}}$

17. Ans (D)

We need inversion of configuration

$\text{Br}_2 / \text{CCl}_4$ gives no reaction

SOBr_2 gives retention

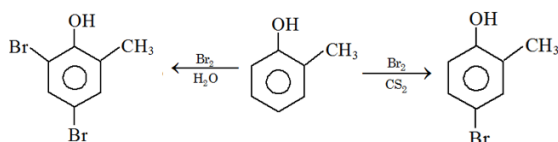
$\text{HBr} / \text{ZnBr}_2$ will give racemisation.

$\text{TsCl} / \text{Pyridine}$ followed by nucleophilic attack by Br^- will give inversion

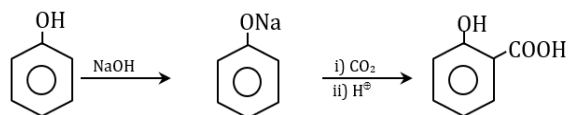
18. Ans (D)

Example of pinacol-pinacolone rearrangement.

19. Ans (C)



20. Ans (A)



In Kolbe's reaction phenoxide ion is treated with carbon dioxide.

PART 2 : CHEMISTRY

SECTION-B

1. Ans (1)

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

$$= 30000 - 3000 \times 10 = 0$$

$$\Delta G^\circ = 0$$

$$-2.303 RT \log K_{eq} = 0$$

$$K_{eq} = 1$$

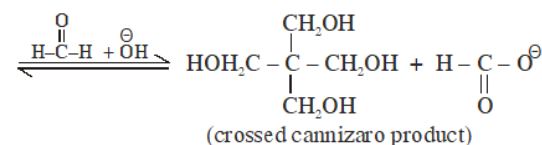
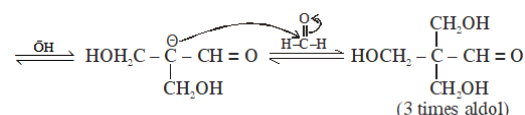
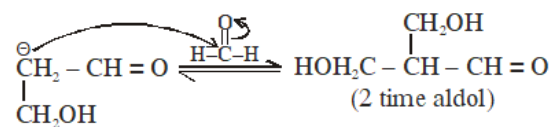
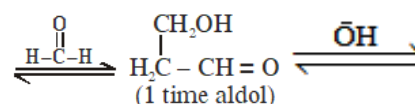
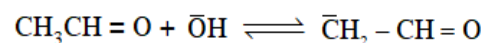
2. Ans (8)

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$

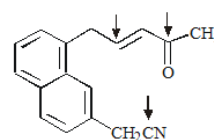
$$21.6 = \Delta H_{\text{H-H}} - 300 (2 \times S^\circ_{\text{H}} - S^\circ_{\text{H}_2})$$

$$\Delta H_{\text{H-H}} = 33.6 \text{ KJ mole}^{-1} = 8 \text{ Kcal mole}^{-1}$$

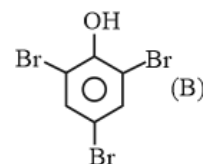
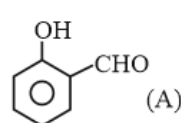
3. Ans (1)



4. Ans (3)



5. Ans (5)



$$x = 2$$

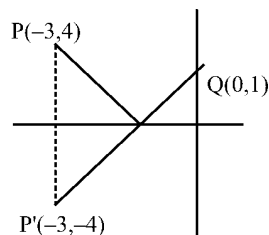
$$y = 3$$

$$x + y = 5$$

PART 3 : MATHEMATICS

SECTION-A

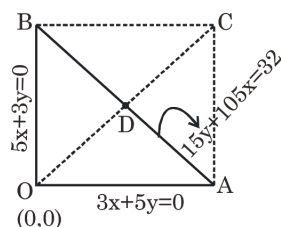
1. **Ans (B)**



Equation of line PQ : $5x - 3y + 3 = 0$

$$\text{put } y = 0, x = -\frac{3}{5}$$

2. **Ans (B)**



On solving $3x + 5y = 0$ and $15y + 105x = 32$

We will get vertex A

$$(3x + 5y = 0) \times 3$$

$$15y + 105x = 32$$

from equations

$$96x = 32 \Rightarrow x = \frac{1}{3}$$

$$A \equiv \left(\frac{1}{3}, -\frac{1}{5}\right)$$

$$\text{Similarly } B \equiv \left(\frac{2}{5}, -\frac{2}{3}\right)$$

$$\Rightarrow D = \left(\frac{\frac{1}{3} + \frac{2}{5}}{2}, \frac{-\frac{1}{5} - \frac{2}{3}}{2}\right) = \left(\frac{11}{30}, \frac{-13}{30}\right)$$

$\Rightarrow$ Equation of OD

$$y = \frac{-13}{11}x \Rightarrow \boxed{11y + 13x = 0}$$

3. **Ans (C)**

Parametric equation of line through A(-1, 3)

$$\text{is } \frac{x+1}{\cos \theta} = \frac{y-3}{\sin \theta} = r$$

$\Rightarrow$ Any point on the line is

$$(r \cos \theta - 1, r \sin \theta + 3)$$

$$\therefore B(AB \cos \theta - 1, AB \sin \theta + 3)$$

$$\& C(AC \cos \theta - 1, AC \sin \theta + 3)$$

lies on $x + y = 4$ & $y = x$ respectively.

$$\Rightarrow AB \cos \theta - 1 + AB \sin \theta + 3 = 4$$

$$\text{or } AB(\cos \theta + \sin \theta) = 2 \quad \dots\dots(1)$$

$$\& AC \cos \theta - 1 = AC \sin \theta + 3$$

$$\text{p } AC(\cos \theta - \sin \theta) = 4 \quad \dots\dots(2)$$

But $AB \cdot AC = 16$

$$\therefore (1) \times (2) \Rightarrow 16(\cos^2 \theta - \sin^2 \theta) = 8$$

$$\text{or } \cos 2\theta = 1/2$$

$$\text{given } \tan \theta = \pm \frac{1}{\sqrt{3}}$$

$$\therefore \left(\frac{m_1 - m_2}{1 + m_1 m_2}\right)^2 = \left(\frac{\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{3}}\right)^2 = 3$$

4. **Ans (A)**

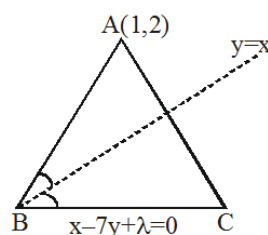
Image of A(1,2) in

$y = x$ lies on

$$x - 7y + \lambda = 0$$

$\Rightarrow (2,1)$ satisfies BC

$$\Rightarrow 2 - 7 + \lambda = 0 = \lambda = 5$$



5. **Ans (D)**

Image of A in $y = x$ lies on BC = $A_1(3, 1)$

Image of A in $y = -2x$ lies on

$$BC = A_{11}(-3, 1)$$

Equation BC, $y = 1$

Ex centre opposite to A $\rightarrow I_1(0, 0)$

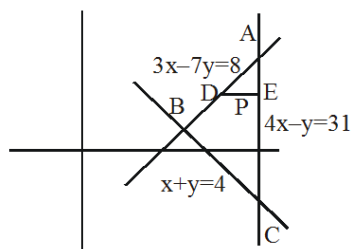
AD $\rightarrow$ equation $y = 3x$

$$\Rightarrow D = \left(\frac{1}{3}, 1\right)$$

$$\frac{AI_1}{I_1 D} = \frac{3}{1} \Rightarrow \frac{AI}{ID} = \frac{3}{1}$$

$$\therefore I = \left(\frac{1}{2}, \frac{3}{2}\right)$$

6. Ans (A)



Equation of DE is $y = 2$

so, point of intersection

$$D\left(\frac{22}{3}, 2\right) \text{ and } E\left(\frac{33}{4}, 2\right)$$

$$\Rightarrow \frac{22}{3} < \lambda < \frac{33}{4}$$

7. Ans (C)

$$x - 3y = q \quad \dots (1)$$

$$ax + 2y = q \quad \dots (2)$$

$$ax + y = r \quad \dots (3)$$

it means equation (1) & (3) are perpendicular

$$m_1 = \frac{1}{3}, m_3 = -a$$

$$m_1 \times m_3 = -1$$

$$a = 3$$

$$\text{It satisfied } a^2 - 9a + 18 = 0$$

8. Ans (C)

Homogenize $x^2 + y^2 - 4x - 2y + 1 = 0$ with line

$$x + \alpha y = 1 \text{ gives}$$

$$x^2 + y^2 - 4x(x + \alpha y) - 2y(x + \alpha y) + (x + \alpha y)^2 = 0$$

For angle to be 90° , we need coeff of x^2 + coeff of $y^2 = 0$

$$(1 - 4 + 1) + (1 - 2\alpha + \alpha^2) = 0$$

$$\Rightarrow \alpha^2 - 2\alpha - 1 = 0$$

$$\text{Sum} = 2$$

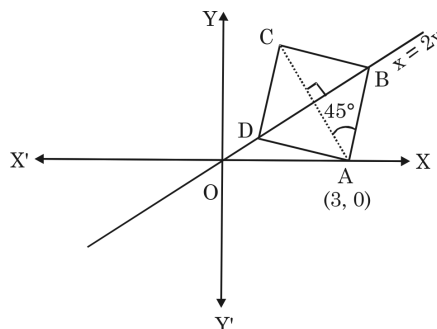
9. Ans (A)

Equation of diagonal AC is

$$y - 0 = -2(x - 3)$$

$$\Rightarrow 2x + y = 6$$

Solving $2x + y = 6$ and $x = 2y$



$$\text{Then, we get } y = \frac{6}{5} \text{ and } x = \frac{12}{5}$$

$$\therefore \text{Centre of square is } \left(\frac{12}{5}, \frac{6}{5}\right)$$

$$\text{Let slope of side AB or AD is } m \text{ then, } \left| \frac{m - (-2)}{1 + m(-2)} \right| = 1$$

$$m = -\frac{1}{3} \quad m = 3$$

$$AB : y - 0 = 3(x - 3)$$

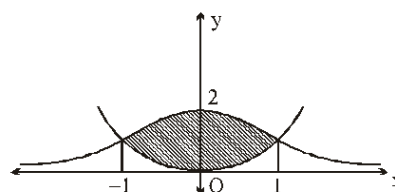
$$AD : y - 0 = -\frac{1}{3}(x - 3)$$

10. Ans (B)

$$\text{Required area} = \int_{-1}^1 \left(\frac{2}{1+x^2} - x^2 \right) dx$$

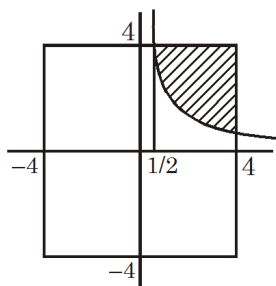
$$= 2 \int_0^1 \left(\frac{2}{1+x^2} - x^2 \right) dx$$

$$= 2 \left(2 \tan^{-1} x - \frac{x^3}{3} \right)_0^1 = 2 \left(\frac{\pi}{2} - \frac{1}{3} \right) = \pi - \frac{2}{3}$$



$$\therefore \left[\pi - \frac{2}{3} \right] = 2$$

11. Ans (A)



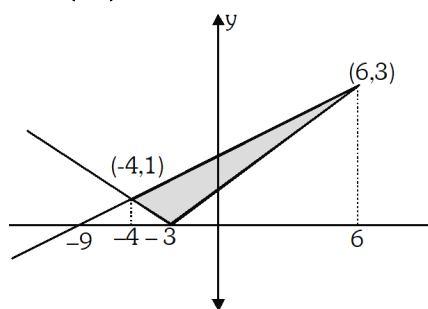
$$|x| \leq 4 \text{ and } |y| \leq 4$$

Area of shaded portion

$$\Delta = \int_{1/2}^4 \frac{2}{x} dx \Rightarrow [2 \ln x]_{1/2}^4 \Rightarrow 2 [\ln 4 + \ln 2] = \ln 64$$

$$\text{Shaded region} = 14 - \ln 64$$

12. Ans (B)



Required Area

$$\frac{1}{2} \times 10 \times 4 - \left[\int_{-4}^{-3} \sqrt{-x-3} dx + \int_{-3}^6 \sqrt{x+3} dx \right]$$

$$20 - \left[\frac{2}{3} [(-x-3)^{3/2}]_{-4}^{-3} + \frac{2}{3} [(x+3)^{3/2}]_{-3}^6 \right]$$

$$20 - \left[\frac{2}{3} [0 - 1] + \frac{2}{3} [9^{3/2}] \right]$$

$$20 - \left[\frac{2}{3} + \frac{2}{3} \times 3^3 \right]$$

$$20 - \left[\frac{2}{3} + 18 \right]$$

$$= 2 - \frac{2}{3} = \frac{4}{3}$$

13. Ans (D)

$$y^2 = 4x; \quad x^2 = 4y$$

$$x = 4; \quad y = 4$$

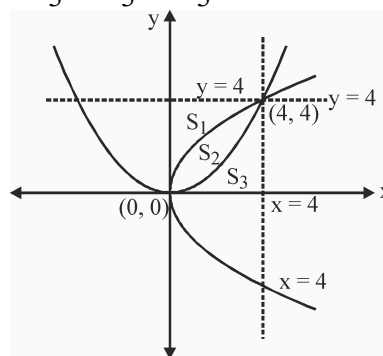
$$x = 0; \quad y = 0$$

$$S_3 = \int_0^4 \frac{x^2}{4} dx = \frac{x^3}{3 \cdot 4} = \frac{64}{3 \cdot 4} = \frac{16}{3}$$

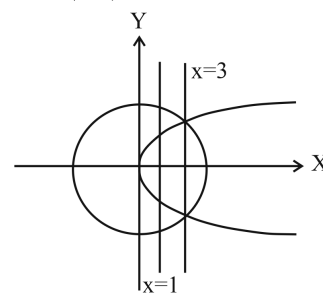
$$S_1 = \int_0^4 \frac{y^2}{4} dy = \frac{y^3}{12} = \frac{64}{12} = \frac{16}{3}$$

$$S_2 = \int_0^4 \left[2\sqrt{x} - \frac{x^2}{4} \right] dx = 2 \cdot \frac{2}{3} x^{3/2} \Big|_0^4 - \frac{x^3}{12} \Big|_0^4$$

$$= \frac{32}{3} - \frac{16}{3} = \frac{16}{3} \Rightarrow S_1 = S_2 = S_3$$



14. Ans (D)



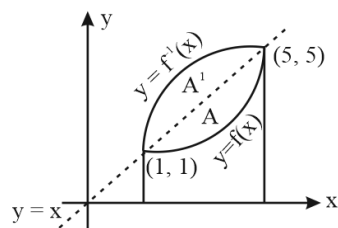
$$\text{area } 2 \int_1^3 2\sqrt{x} dx + 2 \int_3^{\sqrt{21}} \sqrt{21-x^2} dx$$

$$\Delta = \frac{8}{3} (3\sqrt{3} - 1) + 21 \sin^{-1} \left(\frac{2}{\sqrt{7}} \right) - 6\sqrt{3}$$

$$\frac{1}{2} \left(\Delta - 21 \sin^{-1} \left(\frac{2}{\sqrt{7}} \right) \right) = \frac{2\sqrt{3} - \frac{8}{3}}{2}$$

$$= \sqrt{3} - \frac{4}{3}$$

15. Ans (B)



$$A = \left(4 + \frac{1}{2} \cdot 4 \cdot 4\right) - 8$$

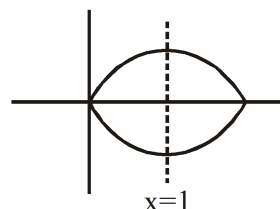
$$= 4 = A^1$$

$$\text{Required area} = 8 + A + A^1$$

$$= 8 + 8 = 16$$

16. Ans (D)

$$y = x\sqrt{2-x}$$



$$2 \int_0^1 x(\sqrt{2-x}) dx$$

$$\text{Let } 2-x = t^2$$

$$dx = -2t dt$$

$$2 \int_{\sqrt{2}}^1 (2-t^2)t(-2t) dt$$

$$= 4 \int_{\sqrt{2}}^1 (2t^2 - t^4) dt$$

$$4 \left(\frac{2t^3}{3} - \frac{t^5}{5} \right) \Big|_{\sqrt{2}}^1$$

$$4 \left(\frac{4\sqrt{2}}{3} - \frac{4\sqrt{2}}{5} \right) - \left(\frac{2}{3} - \frac{1}{5} \right)$$

$$4 \left(\frac{8\sqrt{2}}{15} - \frac{7}{15} \right)$$

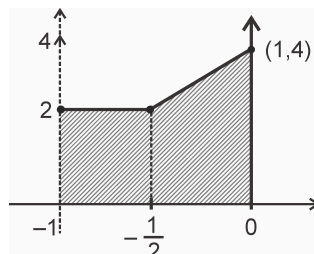
$$\frac{4(8\sqrt{2}-7)}{15}$$

17. Ans (B)

$$f(x) = |2\{x\} - 1| + |2\{x\} + 1|$$

$$|2\{x\} - 1| + |2\{x\} + 1|$$

$$= \begin{cases} 2 & 0 \leq \{x\} < \frac{1}{2} \\ 4\{x\} & \frac{1}{2} \leq \{x\} < 1 \end{cases}$$



$$\text{Required Area} = 2 \times \frac{1}{2} + \frac{1}{2} \times (4+2) \times \frac{1}{2}$$

$$= 1 + \frac{3}{2} = \frac{5}{2} \text{ square units}$$

18. Ans (D)

$$\frac{dy}{dx} + \frac{2^{x-y}(2^y-1)}{2^x-1} = 0$$

$$x, y > 0, y(1) = 1, y(2) = ?$$

$$\frac{dy}{dx} = -\frac{2^x(2^y-1)}{2^y(2^x-1)}$$

$$\int \frac{2^y}{2^y-1} dy = - \int \frac{2^x}{2^x-1} dx$$

$$\frac{1}{\ln 2} \int \frac{2^y \ln 2}{2^y-1} dy = - \frac{1}{\ln 2} \int \frac{2^x \ln 2}{2^x-1} dx$$

$$\frac{1}{\ln 2} \ln|2^y-1| = \frac{-1}{\ln 2} \ln|2^x-1| + C$$

$$\text{At } x=1, y=1$$

$$\text{Putting this values in above relation we get } C=0$$

$$\ln|2^y-1| + \ln|2^x-1| = 0$$

$$(2^x-1)(2^y-1) = 1$$

$$2^y-1 = \frac{1}{2^x-1}$$

$$\text{At } x=2$$

$$2^y = \frac{1}{3} + 1 = \frac{4}{3}$$

$$y = \log_2 \frac{4}{3} = \log_2 4 - \log_2 3 = 2 - \log_2 3$$

19. Ans (B)

Let, $y = tx$

$$\frac{dy}{dx} = t + x \frac{dt}{dx}$$

$$\therefore tx \left(t + x \frac{dt}{dx} \right) = x \left(t^2 + \frac{\phi(t^2)}{\phi'(t^2)} \right)$$

$$t^2 + xt \frac{dt}{dx} = t^2 + \frac{\phi(t^2)}{\phi'(t^2)} \quad \int \frac{t\phi'(t^2)}{\phi(t^2)} dt = \int \frac{dx}{x}$$

$$\text{Let } \phi(t^2) = p \quad \therefore \phi'(t^2) 2t dt = dp$$

$$\Rightarrow \int \frac{dy}{2p} = \int \frac{dx}{x} \quad \frac{1}{2} \ln \phi(t^2) = \ln x + \ln c$$

$$\phi \left(\frac{y^2}{x^2} \right) = kx^2, \phi(1) = k$$

$$\phi(t^2) = x^2 k \phi \left(\frac{y^2}{x^2} \right) = 4\phi(1)$$

20. Ans (C)

$$\frac{dy}{dx} = \frac{y(x+y^3)}{x(y^3-x)}$$

$$\Rightarrow dy(xy^3 - x^2) = (xy + y^4)dx$$

$$\Rightarrow y^3(xdy - ydx) = x(ydx + xdy)$$

$$\Rightarrow x^2 y^3 d\left(\frac{y}{x}\right) = x d(xy)$$

$$\Rightarrow \frac{y}{x} d\left(\frac{y}{x}\right) = \frac{d(xy)}{(xy)^2}$$

$$\Rightarrow \frac{1}{2} \left(\frac{y}{x} \right)^2 = -\frac{1}{xy} + c$$

Passes through (4, -2) $\therefore c = 0$

$$\text{So } y^3 = -2x$$

PART 3 : MATHEMATICS

SECTION-B

1. Ans (1)

$$e^x(1 + ye^x)(dy + ydx) = 2x dx$$

$$\Rightarrow (1 + ye^x)(e^x dy + ye^x dx) = 2x dx$$

$$\Rightarrow \frac{(1 + ye^x)^2}{2} = x^2 + c$$

$$\Rightarrow ye^x + 1 = \sqrt{2x^2 + 4}$$

$$\therefore f(-1) = (\sqrt{6} - 1)e$$

2. Ans (2)

$$(x \cos x)dy + (x y \sin x + y \cos x - 1)dx = 0,$$

$$0 < x < \frac{\pi}{2}$$

$$\frac{dy}{dx} + \left(\frac{x \sin x + \cos x}{x \cos x} \right) y = \frac{1}{x \cos x}$$

$$\text{IF} = x \sec x$$

$$y \cdot x \sec x = \int \frac{x \sec x}{x \cos x} dx = \tan x + c$$

$$\text{Since } y \left(\frac{\pi}{3} \right) = \frac{3\sqrt{3}}{\pi}$$

$$\text{Hence } c = \sqrt{3}$$

$$\text{Hence } \left| \frac{\pi}{6} y \left(\frac{\pi}{6} \right) + y' \left(\frac{\pi}{6} \right) \right| = |-2| = 2$$

3. Ans (6)

$$\frac{dy}{dx} + \frac{4x}{(x^2 - 1)} y = \frac{x + 2}{(x^2 - 1)^{\frac{5}{2}}}, x > 1$$

$$\text{I.F.} = e^{\int \frac{4x}{x^2 - 1} dx}$$

$$\text{I.F.} = (x^2 - 1)^2$$

$$\Rightarrow d(y \cdot (x^2 - 1)^2) = \frac{x + 2}{(x^2 - 1)^{\frac{5}{2}}} \cdot (x^2 - 1)^2$$

$$\Rightarrow \int d(y \cdot (x^2 - 1)^2) = \int \frac{x + 2}{(x^2 - 1)^{\frac{1}{2}}} dx \quad \dots(1)$$

$$y(x^2 - 1)^2 = \sqrt{x^2 - 1} + 2 \ln(x + \sqrt{x^2 - 1}) + C$$

$$\Rightarrow C = -\sqrt{3}$$

$$\text{So } (x^2 - 1)^2 = \sqrt{x^2 - 1} + 2 \ln(x + \sqrt{x^2 - 1}) - \sqrt{3}$$

$$\Rightarrow \alpha\beta\gamma = 6$$

4. Ans (0)

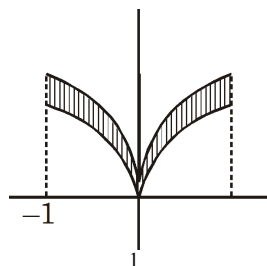
$$f(x) = \sin^{-1} \left(\frac{2|x|}{1+x^2} \right) + \sin^{-1} \left(\frac{|x|}{\sqrt{1+x^2}} \right)$$

Put $|x| = \tan \theta$, $\theta \in \left[0, \frac{\pi}{2}\right)$

$$\therefore f(x) = \sin^{-1}(\sin 2\theta) + \sin^{-1}(\sin \theta)$$

$$= \begin{cases} 2\theta + \theta, & \theta \in \left[0, \frac{\pi}{4}\right] \\ \pi - 2\theta + \theta & \theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \end{cases}$$

$$\therefore f(x) = 3\tan^{-1}|x|, \quad x \in [-1, 1]$$



$$\text{Area} = 2 \int_0^1 (3\tan^{-1}x - \tan^{-1}x) dx$$

$$= 4 \int_0^1 \tan^{-1}x dx$$

$$= \pi - \ln 4$$

$$\therefore [b-a] = [4-\pi] = 0$$

5. Ans (2)

P is intersection point of line AB & $2x + 3y + 1 = 0$

$$\therefore P(7, -5)$$